John Ogheneruese Fiewor

AI Engineer | Founder, GradrAI | AI in Education Researcher
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Research Profile

AI engineer and founder working on trustworthy AI-assisted assessment for African classrooms. My work focuses on end-to-end assessment pipelines: paper-based exam digitisation, rubric-grounded grading, teacher-in-the-loop review, code-switched oral verification, offline/local execution for low-connectivity settings, and safety controls for multi-agent grading systems. I am especially interested in systems that preserve teacher authority while making assessment more scalable, auditable, and accessible.

Research Interests

- Artificial Intelligence in Education (AIEd)
- AI-assisted grading and feedback
- Teacher-in-the-loop and human-in-the-loop assessment systems
- Multilingual and code-switched educational NLP/ASR for African classrooms
- Multi-agent safety, validation, and auditability in grading pipelines
- Psychometrics, grading validity, and responsible educational AI
- Verifiable credentials and trustworthy educational achievement records

Education

Crawford University, Igbesa, Nigeria
M.Sc. Computer Science and Technology — GPA: 4.18 / 5.0
Jan 2021 – Oct 2022

Crawford University, Igbesa, Nigeria
B.Sc. Computer Science and Technology — Second Class Honours (Upper Division); GPA: 4.26 / 5.0
Nov 2014 – Jul 2018

Publications

Peer-Reviewed Workshop Papers

J. O. Fiewor. "Teacher-in-the-Loop AI Assessment for African Classrooms: Paper-Based Grading, Offline Exams, and Code-Switched Oral Verification." *NeurIPS 2026 Workshop on Global South AI*, 2026. Sole author. Accepted 23 Sept 2026. Submission 191. OpenReview: openreview.net/forum?id=IlKAI77Iwm (public release pending).

- Sole-authored study from GradrAI deployment work on paper-based grading under teacher review, code-switched oral assessment, and offline exam workflows.
- Reports a 60-script shadow-mode grading study; 42/60 and 43/60 scripts were routed to teacher review under two grading conditions.
- Evaluates code-switched classroom speech and African benchmark speech; language-hypothesis routing reduced mean WER from 0.63 to 0.15 and the downstream score gap from 2.9 to 1.3 marks out of 10.
- Includes an offline path for topic extraction, exam generation, and essay grading on teacher hardware using local model execution, schema-constrained decoding, recovery caching, and store-and-forward sync.

Research and Professional Experience

Founder & Lead AI Engineer, GradrAI — Lagos, Nigeria

Nov 2024 – Present | gradrai.com

- Founded and lead GradrAI, an AI-assisted assessment platform for automating paper-based exam grading workflows in higher-education settings.
- Architected the grading pipeline from raw document ingestion through text extraction, rubric interpretation, scoring, validation, teacher review routing, feedback generation, and analytics.
- Built the initial OCR-and-rule-based grading system using Microsoft Azure Cognitive Services, then led the transition to LLM-driven grading with Google Gemini on Vertex AI.
- Designed teacher-in-the-loop workflows so automated scores remain reviewable, auditable, and overrideable rather than replacing instructor judgment.
- Conducted shadow-mode evaluations on 60 paper-based scripts to compare grading conditions, identify invalid intermediate outputs, and measure the difference between final score validity and model self-confidence.
- Built code-switched oral-verification experiments for English mixed with Nigerian Pidgin, Yoruba, Swahili, and isiZulu, connecting speech recognition errors to downstream assessment effects.
- Developed an offline/local execution path for low-connectivity classrooms, using locally served models, schema-constrained decoding, recovery caching, and store-and-forward synchronization.
- Shipped the grading engine as a multi-agent system built with Google Agent Development Kit (ADK) and deployed on Vertex AI Agent Engine: sequential pipelines of specialised agents for preprocessing, rubric-based grading, referee cross-checking that routes low-confidence results to teacher review, weakness detection, personalised practice generation, and result aggregation, with MCP tools for database and storage access.

- **Technologies:** Google Gemini, Vertex AI, Google ADK, MCP, Microsoft Azure Cognitive Services, GCP, AWS, React, Node.js, Express, MongoDB, Mongoose.

Full-Stack Developer, VerifyEd — Hampshire, United Kingdom (Remote)

Feb 2025 – Present

Digital credentialing platform for issuing, rendering, sharing, and verifying educational credentials.

- Build and maintain issuer-facing and recipient-facing systems for educational credential issuance and verification across client and administrative interfaces.
- Shipped features improving credential portability and recipient control over presentation and sharing, including landscape-optimised certificate downloads and one-click sharing to professional networks.
- Contributed to the verification surface of educational achievement: how a credential is rendered, distributed, indexed, shared, and trusted by third parties once it leaves the issuing institution.
- Improved server-side rendering so credential pages are crawled more reliably and shared credential links appear correctly on LinkedIn and other platforms.
- Resolved production incidents affecting certificate verification, including XRPL ledger congestion and timeout scenarios.
- Designed and executed a custom script to recover unused XRP from dormant or invalid wallets, saving the company over $1,000.
- Built internal onboarding tooling for vetted customers and organisations, reducing fraudulent sign-ups and improving post-call onboarding flow.
- **Technologies:** Angular, Node.js, Express, PostgreSQL, Sequelize, AWS, Webflow.

Full-Stack Developer, Consolidated Hallmark Insurance — Lagos, Nigeria

Jan 2023 – Jun 2025

- Contributed to a unified internal API that automated core insurance business processes and improved interoperability across teams.
- Played a key role in insourcing the travel insurance portal, moving the product from third-party dependency to an internal platform with lower operational cost and faster iteration.
- Led backend and frontend enhancements that reduced the customer purchase journey from seven steps to three, increasing adoption and improving usability.
- Designed and implemented group, bulk, and Buy Now Pay Later purchase workflows for individuals, organisations, families, and sales agents.
- Temporarily led the frontend engineering team building a modern replacement for a legacy internal insurance operations application.
- Helped establish frontend architecture, CI/CD practices, codebase organisation, and team collaboration patterns.
- **Technologies:** React, JavaScript, Sass, Laravel, PHP, C#.NET, MySQL, SQL Server.

Junior Frontend Developer, HelloHub — London, United Kingdom (Remote)

Feb 2022 – Oct 2022

- Improved mobile responsiveness of the Admin Dashboard by translating Figma designs into production-ready React components.
- Wrote test-driven frontend code and refactored existing components to TypeScript, improving reliability and maintainability.
- Worked with component-driven development, React Testing Library, Jest, Storybook, and clean architecture practices.
- **Technologies:** React, TypeScript, JavaScript, Styled Components, React Testing Library, Jest, Storybook.js.

Frontend Web Development Intern, Zuri — Nigeria (Remote)

Mar 2021 – Jul 2021

- Developed a lease-management web application during a structured remote internship.
- **Technologies:** HTML, CSS, Gulp, Agile methodology.

Teaching and Mentoring Experience

Google Cloud Platform Mentor, Google Africa Developer Scholarship
Jun 2021 – Dec 2021

- Mentored learners working through Google Cloud Platform content and practical cloud-computing exercises.

High School Computer Science Teacher, Apostolic Faith Secondary School
Jan 2019 – Nov 2019

- Taught secondary-school computer science and supported students in foundational computing concepts and practical digital-skills development.

Selected Projects

GradrAI

gradrai.com

- End-to-end AI-assisted assessment platform for paper-based examinations, including document ingestion, grading, feedback, teacher review, analytics, and offline execution.

MarkFlow

ican-grader.vercel.app

- Workflow-management platform for ICAN script marking, supporting examiner assignment, online manual marking, administrative review, and oversight.

Awards and Honors

- **UNDP Nigeria and FSD Africa Innovation Challenge** — Winner, 2025, for work with RapidMedicare (rapidmed.care).
- **Google Gemini National #BuildWithAI Hackathon** — 3rd Place, 2024. Built the initial GradrAI MVP in three days using Google Gemini; independently implemented the frontend in under four hours after a teammate became unavailable and contributed the AI-driven backend component.
- **MongoDB World Hackathon** — 2nd Place, 600+ global participants, 2022. Co-developed "Good News," a web platform surfacing positive news stories from around the world.
- **IBM Call for Code for Racial Justice** — Open-source contributor and Hacktoberfest Demo Day speaker, 2021–2022. Contributed to the Truth Loop project, especially internationalisation and accessibility.

Certifications

- Python Coder — Kaggle, 2025–Present.
- Problem Solving (Basic) — HackerRank, 2025–Present.
- Machine Learning Specialization — DeepLearning.AI and Stanford University, 2024–Present.
- Microsoft Azure AI Engineer Associate, 2021–2024.
- Microsoft Azure Fundamentals, 2023–Present.
- IBM Carbon Design System Developer Essentials — React, 2021–Present.
- Google Cloud Certificate of Course Completion, 2021–Present.
- freeCodeCamp Responsive Web Design Certificate, 2020–Present.

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, C#, HTML, CSS, Sass, SQL, Bash.
- **AI and Agentic Systems:** LLM-assisted grading, prompt engineering, rubric-grounded evaluation, multi-agent pipelines, MCP, Google ADK, Google Gemini, Vertex AI, Azure AI Services.
- **Frameworks and Libraries:** React, Next.js, Angular, Node.js, Express, Redux, .NET Core, Mongoose, Sequelize, Drizzle ORM.
- **Databases:** PostgreSQL, MongoDB, MySQL, SQL Server.
- **Cloud and DevOps:** Google Cloud Platform, AWS, Microsoft Azure, Git, CI/CD, REST APIs.
- **Testing and Engineering Practices:** Jest, React Testing Library, Storybook.js, Agile methodology, clean architecture.

Languages

- English — Native or bilingual proficiency.
- French — Limited working proficiency.

References

Available upon request.